

Patient



Sex: Male

Case Number: TN25-150648

Diagnosis: Non-small cell carcinoma

Specimen Information

Primary Tumor Site: Upper lobe, lung

Specimen Site: Brain, NOS

Specimen ID: S25-07891-A1

Specimen Collected: 19-Mar-2025

Test Report Date: 07-Apr-2025

Ordered By

Reema A. Patel, MD

UK HealthCare - Albert B Chandler

Hospital - Markey Cancer Center

800 Rose Street, Whitney Hendrickson
Building

Lexington, KY 40536 USA

(469) 233 9663

Results with Therapy Associations

BIOMARKER	METHOD	ANALYTE	RESULT	THERAPY ASSOCIATION		BIOMARKER LEVEL*
PD-L1 (22c3)	IHC	Protein	Positive, TPS: 100%	BENEFIT	cemiplimab, pembrolizumab	Level 1
PD-L1 (28-8)	IHC	Protein	Positive 2+, 100%	BENEFIT	nivolumab/ipilimumab combination	Level 1
PD-L1 (SP142)	IHC	Protein	Positive, TC: 1+, 50%	BENEFIT	atezolizumab (metastatic)	Level 1
PD-L1 (SP263)	IHC	Protein	Positive, TC: 2+, 100%	BENEFIT	atezolizumab (adjuvant), cemiplimab	Level 1
ERBB2 (Her2/Neu)	IHC	Protein	Positive Score 3+	BENEFIT	fam-trastuzumab deruxtecan-nxki	Level 2
TMB	Seq	DNA-Tumor	High, 29 mut/Mb	BENEFIT	pembrolizumab	Level 2
ALK	IHC	Protein	Negative 0	LACK OF BENEFIT	alectinib, ceritinib, crizotinib, lorlatinib	Level 1
					brigatinib	Level 2
	Seq	RNA-Tumor	Fusion Not Detected		alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	Level 2
BRAF	Seq	DNA-Tumor	Mutation Not Detected	LACK OF BENEFIT	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	Level 2
EGFR	Seq	DNA-Tumor	Mutation Not Detected	LACK OF BENEFIT	afatinib, dacomitinib, erlotinib, gefitinib, osimertinib	Level 2
KRAS	Seq	DNA-Tumor	Mutation Not Detected	LACK OF BENEFIT	adagrasib, sotorasib	Level 2
MET	Seq	RNA-Tumor	Variant Transcript Not Detected	LACK OF BENEFIT	capmatinib, tepotinib	Level 2
					crizotinib	Level 2
RET	Seq	RNA-Tumor	Fusion Not Detected	LACK OF BENEFIT	pralsetinib, selpercatinib	Level 2
ROS1	Seq	RNA-Tumor	Fusion Not Detected	LACK OF BENEFIT	ceritinib, crizotinib, entrectinib, lorlatinib, repotrectinib	Level 2

* Biomarker reporting classification: Level 1: MI Cancer Seek™ or other companion diagnostic (CDx) performed as part of professional services; Level 2: Strong evidence of clinical significance or is endorsed by standard clinical guidelines; Level 3: Potential clinical significance. Bolded benefit therapies, if present, highlight the most clinically significant findings.

The selection of any, all, or none of the matched therapies resides solely with the discretion of the treating physician. Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, the FDA prescribing information for any therapeutic, and in accordance with the applicable standard of care. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. All trademarks and registered trademarks are the property of their respective owners.

Important Note

This content is provided as part of the professional services. For MI Cancer Seek[™] results, see CDx Associated Findings.

This case was evaluated using a presumed primary site/lineage of LUNG per the clinical history of a lung mass noted in the submitted pathology report. The MI GPSai (Genomic Probability Score) is an algorithm that uses a tumor's molecular signature to provide a differential diagnosis as to a tumor's site of origin and in this case raises the possibility of a GERM CELL TUMOR (Probability 92%), among others including LUNG (Probability 3%). The outside immunohistochemistry evaluation is noted (CK7 and p40 positive, TTF-1 negative). Additional IHC stains performed at Caris show GATA3 diffusely positive, SALL4 variably positive, and inhibin weakly positive in syncytiotrophoblast-like cells. Additionally, HPL and HCG expression levels are increased as determined by sequencing (internal findings). These features raise the possibility of choriocarcinoma; interestingly, however, there is also a mutational pattern by whole exome sequencing that is consistent with tobacco-associated DNA damage, suggesting a primary lung tumor. Together these features raise the possibility of a rare primary pulmonary choriocarcinoma. Of note, a primary mediastinal germ cell tumor would not be expected to show evidence of tobacco-associated DNA damage. Correlation with all available clinical, radiographic, and pathologic information is recommended.

TMB-High status should only be used to guide pembrolizumab treatment when no satisfactory alternative treatment options are available.

For information on Caris GPSai[™], please refer to relevant section of the report (Additional Results Page 1).

Cancer-Type Relevant Biomarkers

Biomarker	Method	Analyte	Result
ERBB2 (Her2/Neu)	CNA-Seq	DNA-Tumor	Amplified
	Seq	DNA-Tumor	Mutation Not Detected
TP53	Seq	DNA-Tumor	Pathogenic Variant Exon 4 p.G117fs
MSI	Seq	DNA-Tumor	Stable
NTRK1/2/3	Seq	RNA-Tumor	Fusion Not Detected
ALK	Seq	DNA-Tumor	Mutation Not Detected
BRAF	Seq	RNA-Tumor	Fusion Not Detected
FGFR3	Seq	RNA-Tumor	Fusion Not Detected
KEAP1	Seq	DNA-Tumor	Mutation Not Detected
MET	CNA-Seq	DNA-Tumor	Amplification Not Detected
	IHC	Protein	Negative 2+, 5%
	Seq	DNA-Tumor	Variant of Uncertain Significance Exon 3 p.R413H

Biomarker	Method	Analyte	Result
MTAP	CNA-Seq	DNA-Tumor	Deletion Not Detected
NFE2L2	Seq	DNA-Tumor	Mutation Not Detected
NRG1	Seq	RNA-Tumor	Fusion Not Detected
PD-L1 (SP142)	IHC	Protein	Negative, IC: 0%
RB1	Seq	DNA-Tumor	Mutation Not Detected
RET	Seq	DNA-Tumor	Mutation Not Detected
STK11	Seq	DNA-Tumor	Mutation Not Detected

Genomic Signatures

Biomarker	Method	Analyte	Result
Microsatellite Instability (MSI)	Seq	DNA-Tumor	Stable
Tumor Mutational Burden (TMB)	Seq	DNA-Tumor	Result: High  Low 10 High
Genomic Loss of Heterozygosity (LOH)	Seq	DNA-Tumor	High - 18% of tested genomic segments exhibited LOH (assay threshold is $\geq 16\%$)

Genes Tested with Pathogenic or Likely Pathogenic Alterations

Gene	Method	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %
BCL2	CNA-Seq	DNA-Tumor	Deleted	-	-	-	-
DACH1	Seq	DNA-Tumor	Pathogenic Variant	c.1436-2A>T	6	c.1436-2A>T	54
ERBB2 (Her2/ Neu)	CNA-Seq	DNA-Tumor	Amplified	-	-	-	-
NOTCH1	Seq	DNA-Tumor	Pathogenic Variant	p.E242*	4	c.724G>T	72
SMAD4	CNA-Seq	DNA-Tumor	Deleted	-	-	-	-
SMARCA4	Seq	DNA-Tumor	Pathogenic Variant	p.E1526*	33	c.4576G>T	68
TP53	Seq	DNA-Tumor	Pathogenic Variant	p.G117fs	4	c.348dupT	76

Unclassified alterations for DNA and RNA sequencing can be found in the MI Portal.
Formal nucleotide nomenclature and gene reference sequences can be found in the Appendix of this report.

Genes Tested with Variants of Uncertain Significance

Gene	Method	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %
DACH1	Seq	DNA-Tumor	Variant of Uncertain Significance	p.H510N	6	c.1528C>A	15
MET	Seq	DNA-Tumor	Variant of Uncertain Significance	p.R413H	3	c.1238G>A	11
ROS1	Seq	DNA-Tumor	Variant of Uncertain Significance	p.R2096L	40	c.6287_6288 delinsTT	41

Additional Variants of Uncertain Significance can be found in the MI Portal.

Predicted Human Leukocyte Antigen (HLA) Genotype Results

The predicted HLA genotype results are summarized in the table below. Please note that HLA typing via tumor tissue sequencing is not a substitute for histocompatibility testing performed using peripheral blood. It is recommended to confirm the patient's HLA type with an appropriate assay.

Gene	Method	Analyte	Genotype
MHC CLASS I			
HLA-A	Seq	DNA-Tumor	A*02:01, A*03:01
HLA-B	Seq	DNA-Tumor	B*07:02, B*47:01
HLA-C	Seq	DNA-Tumor	C*06:02, C*07:02

HLA genotypes with only one allele are either homozygous or have loss of heterozygosity at that position.

Immunohistochemistry Results

Biomarker	Result	Biomarker	Result
ALK	Negative 0	PD-L1 (28-8)	Positive 2+, 100%
ERBB2 (Her2/Neu)	Positive Score 3+	PD-L1 (SP142)	Negative, IC: 0% Positive, TC: 1+, 50%
MET	Negative 2+, 5%	PD-L1 (SP263)	Positive, TC: 2+, 100%
PD-L1 (22c3)	Positive, TPS: 100%		

Genes Tested with Indeterminate Results by Tumor DNA Sequencing

ATP6AP2	COL2A1	EGLN1	FBXW7	MDH2	PLCB4	PREX2	RAD50	RHEB	SMARCE1	STAG2	TRRAP
B2M	CUL3	EIF1AX	FUBP1	MDM2	PMS1	PRKD1	RASA1	RRAS2	SOS1	SUZ12	WRN
CDC73	DGCR8	EXO1	JAK2	MRE11	POLD3	PRKDC	REST	SMARCA2	SSBP1	TRIM28	YES1
CDK6	EED	FANCL	KIF1B	NPM1	PRDM6	RABL3					

Genes in this table were ruled indeterminate due to low coverage for some or all exons.

Genes Tested with Intermediate CNA Results by Tumor DNA Sequencing

GATA3											
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The results in this report were curated to represent biomarkers most relevant for the submitted cancer type. These include results important for therapeutic decision making, as well as notable alterations in other biomarkers known to be involved in oncogenesis. Additional results, including genes with normal findings, variants of uncertain significance, or unclassified alterations can be found in the MI Portal at miportal.carismolecularintelligence.com. If you do not have an MI Portal account, or need assistance accessing it, please contact Caris Customer Support at (888) 979 8669.

PATIENT:

TN25-150648

PHYSICIAN: REEMA A. PATEL, MD

Notes of Significance

SEE APPENDIX FOR DETAILS

CDx reports are generated through automated bioinformatics processing of a pre-defined list of genes and variants. Pathogenic or likely pathogenic variants identified outside the FDA-approved list will not be included in the CDx report. In addition, downgrading of variants approved by board-eligible/board-certified geneticists and pathologists may have occurred through manual variant annotation carried out by Caris' professional services. To ensure optimal coverage, multiple sequencing runs may be performed for a given case. This may account for some discrepancies observed between CDx reports and professional services.

Clinical Trials Connector™ opportunities based on biomarker expression: 544 Targeted Therapies. See page 6 for details.

Please Note: This case was evaluated using a presumed primary site/lineage of LUNG per the clinical history of a lung mass noted in the submitted pathology report. The MI GPSai (Genomic Probability Score) is an algorithm that uses a tumor's molecular signature to provide a differential diagnosis as to a tumor's site of origin and in this case raises the possibility of a GERM CELL TUMOR (Probability 92%), among others including LUNG (Probability 3%). The outside immunohistochemistry evaluation is noted (CK7 and p40 positive, TTF 1 negative). Additional IHC stains performed at Caris show GATA3 diffusely positive, SALL4 variably positive, and inhibin weakly positive in syncytiotrophoblast like cells. Additionally, HPL and HCG expression levels are increased as determined by sequencing (internal findings). These features raise the possibility of choriocarcinoma; interestingly, however, there is also a mutational pattern by whole exome sequencing that is consistent with tobacco associated DNA damage, suggesting a primary lung tumor. Together these features raise the possibility of a rare primary pulmonary choriocarcinoma. Of note, a primary mediastinal germ cell tumor would not be expected to show evidence of tobacco associated DNA damage. Correlation with all available clinical, radiographic, and pathologic information is recommended.

Specimen Information

Specimen ID: S25 07891 A1

Specimen Collected: 03/19/2025

Specimen Received: 03/26/2025

Testing Initiated: 03/29/2025

Test Ordered*: MI Profile™ (MI Cancer Seek™ + IHCs and Other Tests by Tumor Type)

* If the submitted specimen is inadequate, only a subset of the ordered testing may be reported.

Gross Description: 1 (A) Paraffin Block Client ID(S25 07891 A1) from UK HealthCare Albert B Chandler Hospital, Lexington, KY, with the corresponding pathology report labeled "S25 07891".

Dissection Information: Molecular testing of this specimen was performed after harvesting of targeted tissues with an approved manual microdissection technique. Candidate slides were examined under a microscope and areas containing tumor cells (and separately normal cells, when necessary for testing) were circled. A laboratory technician harvested targeted tissues for extraction from the marked areas using a dissection microscope.

PATIENT: [REDACTED]

TN25-150648

PHYSICIAN: REEMA A. PATEL, MD

Clinical Trials Connector™

The Clinical Trials Connector lists agents that are matched to available clinical trials according to biomarker status. In some instances, older generation agents may still be relevant in the context of new combination strategies and, therefore, will still appear on this report.

Therapeutic agents listed below may or may not be currently FDA approved for the tumor type tested.

Please see <https://clinicaltrials.gov/> for more information.

TARGETED THERAPIES (544)				
Drug Class	Biomarker	Method	Analyte	Investigational Agent(s)
Bi-specific antibody (1)	HLA-A	NGS	DNA-Tumor	IMC-F106C
EZH2 inhibitors (6)	SMARCA4	NGS	DNA-Tumor	CPI-0209, tazemetostat, valemetostat
HDAC inhibitors (2)	NOTCH1	NGS	DNA-Tumor	entinostat, valproic acid
Heat shock pathway inhibitors (1)	SMARCA4	NGS	DNA-Tumor	NXP800
HER2 antibody drug conjugates (27)	ERBB2 (Her2/Neu)	CNA-NGS	DNA-Tumor	ado-trastuzumab emtansine (T-DM1), DB-1303, disitamab vedotin, GQ1001, trastuzumab deruxtecan
	ERBB2 (Her2/Neu)	IHC	Protein	
HER2-targeted therapy (50)	ERBB2 (Her2/Neu)	CNA-NGS	DNA-Tumor	DF1001, pertuzumab, trastuzumab, tucatinib, zanidatamab
	ERBB2 (Her2/Neu)	IHC	Protein	
Immunomodulatory agents (439)	PD-L1	IHC	Protein	atezolizumab, avelumab, balstilimab, cemiplimab, cetrelimab, dostarlimab, durvalumab, envafoimab, ipilimumab, nivolumab, pembrolizumab, retifanlimab, spartalizumab, tislelizumab, toripalimab, tremelimumab, XmAb20717, zimberelimab
	PD-L1 (SP142)	IHC	Protein	
	TMB	NGS	DNA-Tumor	
Pan-HER inhibitors (16)	ERBB2 (Her2/Neu)	CNA-NGS	DNA-Tumor	afatinib, dacomitinib, neratinib, pyrotinib
	ERBB2 (Her2/Neu)	IHC	Protein	
SWI/SNF complex inhibitors (2)	SMARCA4	NGS	DNA-Tumor	PRT3789

() = represents the total number of clinical trials identified by the Clinical Trials Connector for the provided drug class or table.

The Clinical Trials Connector may include trials that enroll patients with additional screening of molecular alterations. In some instances, only specific gene variants may be eligible.

Disclaimer

Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, prescribing information for any therapeutic, and in accordance with the applicable standard of care. Drug associations provided in this report do not guarantee that any particular agent will be effective for the treatment of any patient or for any particular condition. Caris Life Sciences® expressly disclaims and makes no representation or warranty whatsoever relating, directly or indirectly, to the performance of services, including any information provided and/or conclusions drawn from therapies that are included or omitted from this report. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. The selection of therapy, if any, resides solely in the discretion of the treating physician and the tests should not be considered a companion diagnostic.

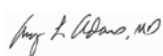
Caris MPI, Inc. d/b/a Caris Life Sciences is certified under the Clinical Laboratory Improvement Amendments (CLIA) as qualified to perform high complexity clinical laboratory testing, including all Caris molecular profiling assays. Individual assays that are available through Caris molecular profiling include both Laboratory Developed Tests (LDT) and U.S. Food and Drug Administration (FDA) approved or cleared tests. In addition, certain tests have been CE marked as a general IVD under the In Vitro Diagnostic Directive (IVDD) 98/79/EC. Offered LDTs were developed and their performance characteristics determined by Caris. Certain tests have not been cleared or approved by the FDA. Caris LDTs are used for clinical purposes. They are not investigational or for research. Caris' CLIA certification number is located at the bottom of each page of this report.

The information presented in the Clinical Trials Connector™ section of this report, if applicable, is compiled from sources believed to be reliable and current. However, the accuracy and completeness of the information provided herein cannot be guaranteed. The clinical trials information present in the biomarker description was compiled from www.clinicaltrials.gov. The contents are to be used only as a guide, and health care providers should employ their best comprehensive judgment in interpreting this information for a particular patient. Specific eligibility criteria for each clinical trial should be reviewed as additional inclusion criteria may apply.

All materials, documents, data, data software, information and/or inventions supplied to customers by or on behalf of Caris or created by either party relating to the services shall be and remain the sole and exclusive property of Caris. Customer shall not use or disclose the information provided by Caris through the services or related reports except in connection with the treatment of the patient for whom the services were ordered and shall not use such property for, or disseminate such property to, any third parties without expressed written consent from Caris. Customer shall deliver all such property to Caris immediately upon demand or upon Caris ceasing to provide the services. The technical and professional component of all testing was performed at the laboratory location displayed in the footer unless otherwise notated in the report.

Caris molecular testing is subject to Caris' intellectual property. Patent www.CarisLifeSciences.com/ip.

Professional Component Performed: Location 13, Phoenix, AZ. Amy Adams, MD, Lab Director.

A handwritten electronic signature in black ink, appearing to read "Amy A. Adams, MD".

Electronic Signature
Amy Adams, MD
07 Apr 2025



(01)00860008613325(21)H53Y5DSXF

MI Cancer Seek™

Patient



Sex: Male

Case Number: TN25-150648

CDx Report Generated: 04/02/2025

Specimen Information

Diagnosis and Tumor Type:

Non-small cell carcinoma, Upper lobe, lung

Specimen Site: Brain, NOS

Specimen ID: S25-07891-A1

Specimen Type: Formalin-fixed paraffin-embedded

Specimen Collected: 03/19/2025

Ordered By

Reema A. Patel, MD

UK HealthCare - Albert B Chandler

Hospital - Markey Cancer Center

CDx Associated Findings

Genomic Findings Detected

FDA-approved Therapeutic Options

Findings did not yield any biomarker results with Companion Diagnostic (CDx) Claims.

EGFR L858R

Not Detected

EGFR exon 19 deletions

Not Detected

Microsatellite Instability (MSI)

Not MSI-H

Tumor Profiling Results

MI Cancer Seek is FDA approved to provide tumor mutation profiling results for previously diagnosed oncology patients with solid tumors.

Other Alterations and Biomarkers Identified

Results reported in this section are not prescriptive or conclusive for labeled use of any specific therapeutic product. Confirmation of tumor mutation status using an FDA approved CDx test is needed for therapeutic use.

Tumor Mutational Burden (TMB)	29 mut/Mb
NOTCH1	E242*
SMARCA4	E1526*
TP53	G117fs

MI Cancer Seek™

Intended Use:

MI Cancer Seek is a next-generation sequencing (NGS) based in vitro diagnostic (IVD) device using total nucleic acid (TNA) isolated from formalin-fixed paraffin-embedded (FFPE) tumor tissue specimens for the detection of single nucleotide variants (SNVs) and insertions and deletions (indels) in 228 genes, microsatellite instability (MSI), tumor mutational burden (TMB) in patients with previously diagnosed solid tumors, and copy number amplification (CNA) in one gene in patients with breast cancer.

MI Cancer Seek is intended as a companion diagnostic to identify patients who may benefit from treatment with the targeted therapies listed in Table 1 below, in accordance with the approved therapeutic product labeling.

Additionally, MI Cancer Seek is intended to provide tumor mutational profiling to be used by qualified healthcare professionals in accordance with professional oncology guidelines for cancer patients with previously diagnosed solid malignant neoplasms. Genomic findings other than those listed in Table 1 are not prescriptive or conclusive for labeled use of any specific therapeutic product.

Table 1. MI Cancer Seek Companion Diagnostic Indications

INDICATION	BIOMARKER	THERAPY
Breast Cancer	<i>PIK3CA</i> (C420R, E542K, E545A, E545D [1635G>T only], E545G, E545K, Q546E, Q546R, H1047L, H1047R, H1047Y)	PIQRAY® (alpelisib)
Colorectal Cancer (CRC)	<i>KRAS</i> wild-type (absence of mutations in exons 2, 3, and 4) and <i>NRAS</i> wild-type (absence of mutations in exons 2, 3, and 4)	VECTIBIX® (panitumumab)
	<i>BRAF</i> V600E	BRAFTOVI® (encorafenib) in combination with ERBITUX® (cetuximab)
Melanoma	<i>BRAF</i> V600E	BRAF Inhibitors approved by FDA*
	<i>BRAF</i> V600E or <i>BRAF</i> V600K	MEKINIST® (trametinib) or BRAF/MEK Inhibitor Combinations approved by FDA*
Non-small cell lung cancer (NSCLC)	<i>EGFR</i> exon 19 deletions and exon 21 L858R alterations	EGFR Tyrosine Kinase Inhibitors approved by FDA*
Solid Tumors	MSI-H	KEYTRUDA® (pembrolizumab), JEMPERLI (dostarlimab-gxly)
Endometrial Carcinoma	Not MSI-H	KEYTRUDA® (pembrolizumab) in combination with LENVIMA® (lenvatinib)

*For the most current information about the therapeutic products in this group, go to:

https://www.fda.gov/medical-devices/in-vitro-diagnostics/list-cleared-or-approved-companion-diagnostic-devices-in-vitro-and-imaging-tools#Group_Labeling

MI Cancer Seek™ is a single-site assay performed at Caris Life Sciences, Phoenix, AZ.

Contraindications:

There are no known contraindications.

Warnings and Precautions:

Biopsy may pose a risk to the patient when archival tissue is not available for use with the assay. The patient's physician should determine whether the patient is a candidate for biopsy.

Limitations:

- For *in vitro* diagnostic use.
- CAUTION: Federal law restricts this device to sale by or on the order of a physician.
- The acceptable preparation method for MI Cancer Seek CDx specimens is FFPE. Other preparations have not been evaluated.
- The test is designed to report out somatic variants and is not intended to report germline variants.
- MI Cancer Seek requires a minimum tumor percentage of 20% for detection of alterations, with tumor content enrichment recommended for specimens with tumor percentage lower than 20%.
- Genomic findings other than those listed in the Companion Diagnostic Indications table are not prescriptive or conclusive for labeled use of any specific therapeutic product. Confirmation of tumor mutation status using an FDA-approved CDx test is needed for therapeutic use.
- A negative result does not rule out the presence of a mutation below the limits of detection of the assay.
- MI Cancer Seek is only approved for use with Caris Life Sciences pre-qualified Illumina NovaSeq 6000 instruments.
- The test is intended to be performed on specific serial number-controlled instruments by Caris Life Sciences.
- MI Cancer Seek does not report TMB for values lower than 3 mut/Mb as the accuracy of TMB values below 3 mut/Mb are unreliable.
- Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all applicable information concerning the patient's condition, such as patient and family history, physical examinations, information from other diagnostic tests, and patient preferences, in accordance with the standard of care in a given community.
- Based on the low positive percent agreement (PPA) in the accuracy study for ERBB2 copy number amplifications (CNAs) in breast cancer patients, this alteration may not be detected. Additional clinical investigation to confirm the presence of ERBB2 CNAs in the breast cancer patient's tumor with another FDA approved or cleared test is strongly recommended.
- Patients with breast cancer whose specimens have intermediate ERBB2 CNA status should be tested with another FDA approved or cleared test to ascertain ERBB2 CNA status in their tumor.
- Test may have reduced sensitivity and may yield false negative results in samples where necrotic tissue is >15%, melanin is >5%, fat cells are >10%.

Test Principle:

MI Cancer Seek is a single-site assay performed at Caris Life Sciences located at 4610 South 44th Place, Phoenix, AZ 85040. The test includes reagents, software, and procedures for testing of total nucleic acid (TNA) from formalin fixed paraffin embedded (FFPE) tumor tissue. The test uses a custom bait panel to measure all coding regions of the exome to detect and report SNVs and indels within 228 genes outlined in Table 2 across solid tumors and amplifications in ERBB2 in patients with breast cancer only. The test also detects MSI (determined from 3,210 genes) and whole exome based TMB in patients with solid tumors.

Table 2. MI Cancer Seek Reportable Gene List for SNVs and indels										
ABL1	BCL9	CDKN2A	ESR1	FLT1	JAK3	MET	NRAS	PPP2R2A	RUNX1	STK11
ACVR1	BCOR	CHEK1	EZH2	FLT3	KDM5C	MITF	NSD1	PRDM1	SDHA	SUFU
AIP	BLM	CHEK2	FANCA	FOXA1	KDM6A	MLH1	NSD2	PRKACA	SDHAF2	TCF7L2
AKT1	BMPR1A	CIC	FANCB	FOXL2	KDR	MLH3	NTHL1	PRKAR1A	SDHB	TERT
AKT2	BRAF	CREBBP	FANCC	FUBP1	KEAP1	MPL	NTRK1	PRKDC	SDHC	TET2
AKT3	BRCA1	CSF1R	FANCD2	GATA3	KIT	MRE11	NTRK2	PTCH1	SDHD	TMEM127
ALK	BRCA2	CTCF	FANCE	GNA11	KLF4	MSH2	NTRK3	PTEN	SETD2	TNFAIP3
AMER1	BRIP1	CTNNA1	FANCF	GNA13	KMT2A	MSH3	PALB2	PTPN11	SF3B1	TNFRSF14
APC	BTB	CTNNA1	FANCG	GNAQ	KMT2C	MSH6	PBRM1	RAC1	SMAD2	TP53
AR	CALR	CXCR4	FANCI	GNAS	KMT2D	MTOR	PDGFRA	RAD50	SMAD4	TRAF7
ARAF	CARD11	CYLD	FANCL	H3F3A	KRAS	MUTYH	PDGFRB	RAD51B	SMARCA4	TSC1
ARID1A	CBFB	DDR2	FANCM	H3F3B	LZTR1	MYC	PIK3CA	RAD51C	SMARCB1	TSC2
ARID2	CCND1	DICER1	FAS	HIST1H3B	MAP2K1	MYCN	PIK3CB	RAD51D	SMARCE1	U2AF1
ASXL1	CCND2	DNMT3A	FAT1	HNF1A	MAP2K2	MYD88	PIK3R1	RAD54L	SMO	VHL
ATM	CCND3	EGFR	FBXW7	HOBX13	MAP2K4	NBN	PIK3R2	RAF1	SOC1	WRN
ATRX	CD79B	EP300	FGFR1	HRAS	MAP3K1	NF1	PIM1	RASA1	SOS1	WT1
AXIN2	CDC73	EPHA2	FGFR2	IDH1	MAPK1	NF2	PMS2	RB1	SPEN	XPO1
B2M	CDH1	ERBB2	FGFR3	IDH2	MAX	NFE2L2	POLD1	RET	SPOP	XRCC1
BAP1	CDK12	ERBB3	FGFR4	IRF4	MED12	NFKBIA	POLE	RHOA	SRC	
BARD1	CDK4	ERBB4	FH	JAK1	MEF2B	NOTCH1	POT1	RNF43	STAG2	
BCL2	CDKN1B	ERCC2	FLCN	JAK2	MEN1	NPM1	PPP2R1A	ROS1	STAT3	

FDA Evidence Levels:

Genomic findings other than those listed in the Intended Use are not prescriptive or conclusive for labeled use of any specific therapeutic product. Test results should be interpreted in the context of pathological evaluation of tumors, treatment history, clinical findings, and other laboratory data. The test report includes genomic findings reported in the following levels (Table 3).

Table 3. Classification Criteria for FDA Evidence Levels

LEVEL	CRITERIA
Level 1	Biomarker is FDA-approved as a companion diagnostic as part of MI Cancer Seek™
Level 2	Cancer alterations that are well-established with strong clinical evidence that the clinician must know according to professional consensus guidelines in the specific tumor type.
Level 3	Cancer alterations with potential clinical significance, e.g., biomarkers deemed useful for directing patients to a clinical trial or simply for informational purposes.
	i. Clinical data such as case reports, single or several case series, or Phase I/II clinical trial data that support the utility of specific biomarker alteration to direct a patient to clinical trials, or ii. Pre-clinical and/or <i>in vitro</i> studies provide structural analysis of the mutation, fusion, or isoform to confirm pathogenicity (tumor-promoting), sensitivity, or resistance through colony forming assays, growth inhibition or drug sensitivity assays, etc.



The Caris GPSai (Caris Genomic Probability Score – Artificial Intelligence) is a cancer-type similarity assessment which compares the characteristics of a patient's tumor against other tumors in the Caris database. Caris GPSai analyzes a tumor's molecular signature and provides the prevalence of that signature in the Caris Life Sciences genomic and transcriptomic database across 90 distinct cancer categories.

Most Likely Tumor Type: Germ Cell Tumor

92%

The predicted probability of each cancer type that can be detected by the product are shown below. The total probabilities of the top level cancer type (blue) total 100%. The subtypes (indented) sum to the total probability predicted for the top level cancer type. Discrepancies between expected sums and displayed sums may be exhibited due to rounding.

0	Adrenal Cortical Carcinoma
1	Bladder/Urinary Tract
1	Urothelial Carcinoma
0	Bladder Adenocarcinoma
0	Bowel
0	Appendiceal Adenocarcinoma
0	Colorectal Adenocarcinoma
0	Small Bowel Carcinoma
0	Breast
0	Metaplastic Breast Cancer
0	Breast Invasive Lobular Carcinoma
0	Breast Invasive Ductal Carcinoma
0	CNS/Brain
0	Diffuse Glioma
0	Meningioma
0	Cervix/Uterine Carcinoma
0	Uterine Serous Carcinoma/Uterine Papillary Serous Carcinoma
0	Uterine Carcinosarcoma/Uterine Malignant Mixed Mullerian Tumor
0	Uterine Clear Cell Carcinoma
0	Cervical Adenocarcinoma
0	Uterine Endometrioid Carcinoma
0	Cutaneous Squamous Cell Carcinoma
3	Esophagus/Stomach
1	Esophageal Squamous Cell Carcinoma
2	Esophagogastric Adenocarcinoma
2	Stomach Adenocarcinoma
0	Adenocarcinoma of the Gastroesophageal Junction
0	Esophageal Adenocarcinoma
92	Germ Cell Tumor
0	Hematological
0	Hepatocellular Carcinoma
0	Kidney
0	Renal Cell Carcinoma
0	Chromophobe Renal Cell Carcinoma
0	Papillary Renal Cell Carcinoma
0	Renal Clear Cell Carcinoma
0	Wilms Tumor
0	Melanoma
0	Mesothelioma
0	Neuroendocrine Neoplasm
0	Well/Moderately-Differentiated Neuroendocrine Tumor
0	Poorly-Differentiated Neuroendocrine Carcinoma
0	Large Cell Neuroendocrine Carcinoma
0	Small Cell Neuroendocrine Carcinoma
0	Paraganglioma/Pheochromocytoma
0	Merkel Cell Carcinoma

3	Non-Small Cell Lung Carcinoma
2	Lung Squamous Cell Carcinoma
1	Lung Adenocarcinoma
0	Orogenital Squamous Cell Carcinoma
0	Ovarian Epithelial Tumor
0	Ovarian Carcinosarcoma/Malignant Mixed Mesodermal Tumor
0	Serous Ovarian/Fallopian Tube/Peritoneal
0	Low-Grade Serous Ovarian/Fallopian Tube/Peritoneal Cancer
0	High-Grade Serous Ovarian/Fallopian Tube/Peritoneal Cancer
0	Mucinous Ovarian Cancer
0	Clear Cell Ovarian Cancer
0	Endometrioid Ovarian Cancer
0	Pancreatobiliary
0	Gallbladder Cancer
0	Pancreatic Adenocarcinoma
0	Cholangiocarcinoma
0	Peripheral Nervous System
0	Schwannoma
0	Neuroblastoma
0	Malignant Peripheral Nerve Sheath Tumor
0	Prostate Adenocarcinoma
0	Salivary Gland Tumor
0	Sex Cord Stromal Tumor
0	Granulosa Cell Tumor
0	Sertoli-Leydig Cell Tumor
0	Soft Tissue/Bone
0	Gastrointestinal Stromal Tumor
0	Leiomyosarcoma
0	Angiosarcoma
0	Rhabdomyosarcoma
0	Synovial Sarcoma
0	Osteosarcoma
0	Liposarcoma
0	Chondrosarcoma
0	Endometrial Stromal Sarcoma
0	Ewing Sarcoma
0	Thymic Carcinoma
0	Thyroid
0	Medullary Thyroid Cancer
0	Hurthle Cell Thyroid Cancer
0	Well-Differentiated Thyroid Cancer
0	Follicular Thyroid Cancer
0	Papillary Thyroid Cancer
0	Anaplastic Thyroid Cancer

PATIENT: [REDACTED]

TN25-150648

PHYSICIAN: REEMA A. PATEL, MD

Gene Expression

Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type
ADORA2A	27	FOLR1	78	NTRK3	54
ALK	18	GPC3	4	PDCD1	22
ASCL1	30	HGF	26	PDCD1LG2	21
ATM	28	HRAS	42	PIK3CA	42
AURKA	98	IGF1R	87	POU3F2	44
BRAF	68	KDM1A	57	PRAME	88
BRCA1	100	KDR	22	PTEN	63
BRCA2	100	KEAP1	9	RB1	1
BRD4	76	KRAS	46	RET	62
CCND1	15	LAG3	43	ROR1	32
CCND2	4	MAGEA4	29	ROR2	5
CCNE1	100	MDM2	63	ROS1	4
CD274	91	MET	22	SLFN11	4
CD276	100	MKI67	99	SRC	66
CDKN2A	40	MSLN	16	SSTR2	9
CEACAM5	4	MTAP	76	SSTR3	76
CLDN18	34	MTOR	10	SSTR5	49
CLDN6	94	MUC1	4	STK11	12
CTLA4	48	MUC16	19	TACSTD2	0
DLL3	84	MYC	21	TGFB1	22
EGFR	74	NECTIN4	21	TOP1	82
EPHA2	2	NEUROD1	20	TP53	20
ERBB2	99	NF1	19	TSC1	90
ERBB3	50	NRAS	100	TSC2	7
FGFR1	72	NRG1	12	VEGFA	3
FGFR2	10	NTRK1	52	XPO1	82
FGFR3	34	NTRK2	46	YAP1	15

Gene Expression of Selected Genes by Whole Transcriptome Sequencing (WTS) Methods:

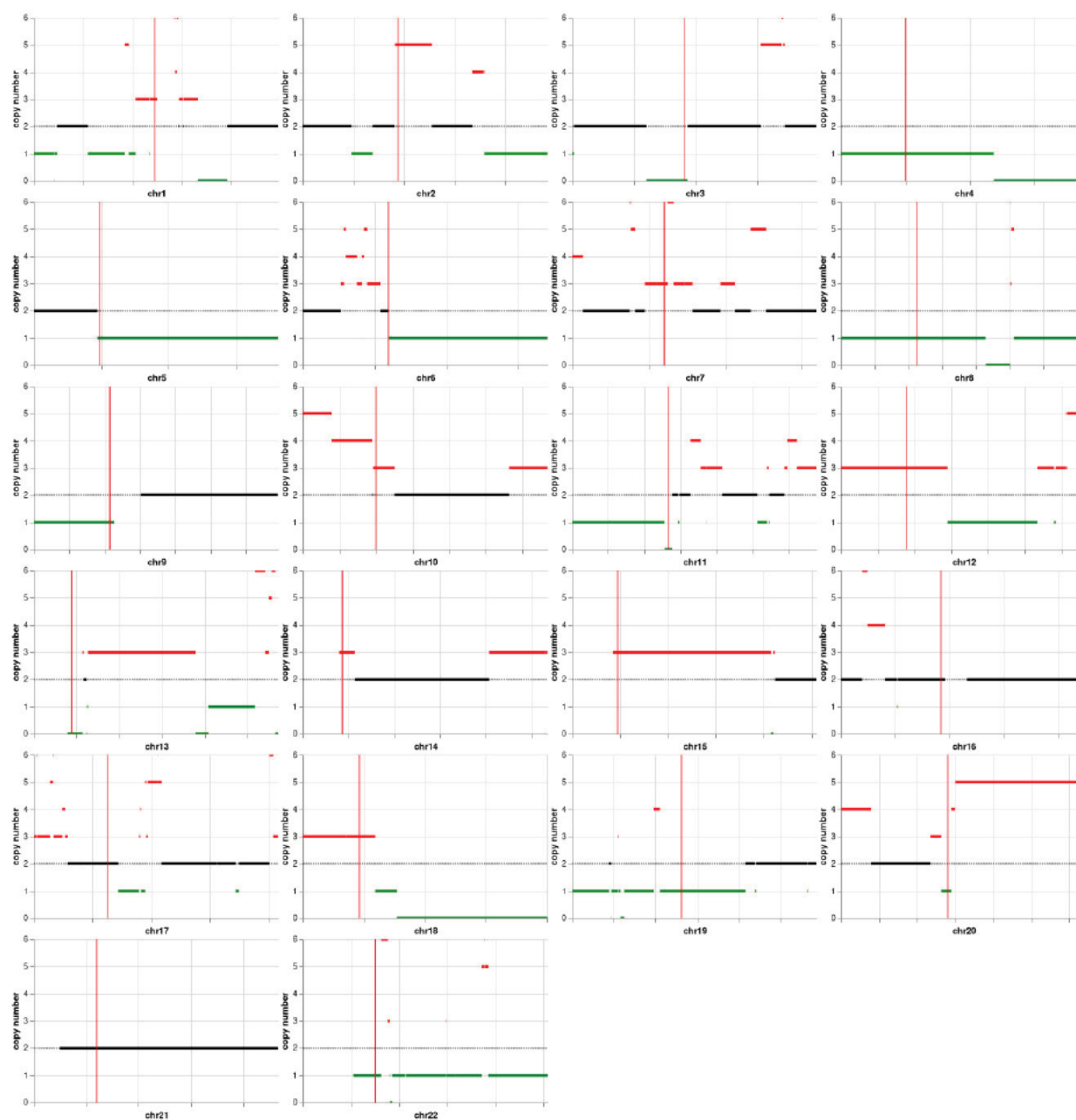
Gene expression is derived from whole transcriptome sequencing. Relative expression of genes are calculated as normalized values using transcripts per Million Molecules or TPM. TPM is presented as a percentile derived by comparison to a distribution of Caris' internal cohort of the tumor-type profiled. Selected genes reported in this section were chosen based on their tumor-type specific relevance for matching to clinical trials, or tumor type subclassification.

PATIENT: [REDACTED]

TN25-150648

PHYSICIAN: REEMA A. PATEL, MD

Karyotype



Karyotyping using Copy Number Analysis by Whole Exome Sequencing (WES) Methods:

Whole exome sequencing in combination with interrogation of single nucleotide polymorphisms (SNPs) tiled throughout the genome, allows for the identification and visualization of cytogenetic aberrations.

Somatic structural variants like whole or partial chromosome duplications or deletions, are important for cancer development and progression, and may identify clinically actionable alterations.

Mutational Analysis by Next-Generation Sequencing (NGS)

TUMOR MUTATIONAL BURDEN	
Mutations / Megabase	Result
29	High

TMB

Tumor Mutational Burden (TMB) is defined as the number of somatic non synonymous mutations per million bases of sequenced DNA in a tumor sample. Tumors with high TMB may increase the number of neoantigens which is hypothesized to increase T cell reactivity and potential for response to immune checkpoint inhibitors. TMB analysis was performed based on next generation sequencing analysis of genomic DNA isolated from a tumor sample.

MICROSATELLITE INSTABILITY ANALYSIS	
Test	Result
MSI	Stable

MSI

Microsatellite instability (MSI) status is a measure of the number of somatic mutations within short, repeated sequences of DNA (microsatellites). MSI High status can indicate that the tumor has a defect in mismatch repair (MMR) abrogating the ability to correct mistakes during DNA replication. Tumors with MSI high status may increase the number of neoantigens which is hypothesized to increase T cell reactivity and potential for response to immune checkpoint inhibitors. Tumor only microsatellite instability status by NGS (MSI NGS) is measured by the direct analysis of known microsatellite regions sequenced in the CMI NGS panel.

GENOMIC LOSS OF HETEROZYGOSITY	
Test	Result
Genomic Loss of Heterozygosity (LOH)	High - 18% of tested genomic segments exhibited LOH (assay threshold is $\geq 16\%$)

LOH

To calculate genomic loss of heterozygosity (LOH), the 22 autosomal chromosomes are split into 552 segments and the LOH of single nucleotide polymorphisms (SNPs) within each segment is calculated. Caris WES data consist of approximately 250k SNPs spread across the genome. SNP alleles with frequencies skewed towards 0 or 100% indicate LOH (heterozygous SNP alleles have a frequency of 50%). The final call of genomic LOH is based on the percentage of all 552 segments with observed LOH.

Additional Next-Generation Sequencing results continued on the next page. >

PATIENT:

TN25-150648

PHYSICIAN: REEMA A. PATEL, MD

Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
DACH1	DNA-Tumor	Pathogenic Variant	c.1436-2A>T	6	c.1436-2A>T	54	NM_080759.5

Interpretation: A pathogenic mutation that disrupts an intron splice site was detected in DACH1.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
DACH1	DNA-Tumor	Variant of Uncertain Significance	p.H510N	6	c.1528C>A	15	NM_080759.5

Interpretation: A variant with no known clinical or functional significance was detected in DACH1.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
MET	DNA-Tumor	Variant of Uncertain Significance	p.R413H	3	c.1238G>A	11	NM_001127500.2

Interpretation: A missense variant of unknown clinical significance in the MET gene was found in the sample.

C-Met is a proto-oncogene that encodes the tyrosine kinase receptor of hepatocyte growth factor (HGF) or scatter factor (SF). c-Met mutation causes aberrant MET signaling in various cancer types including renal papillary, hepatocellular, head and neck squamous, gastric carcinomas and non-small cell lung cancer. Mutations in the juxtamembrane domain (exon 14, 15) results in the constitutive activation and show enhanced tumorigenicity. Germline mutations in c-MET have been associated with hereditary papillary renal cell carcinoma.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
NOTCH1	DNA-Tumor	Pathogenic Variant	p.E242*	4	c.724G>T	72	NM_017617.4

Interpretation: A nonsense (loss of function) mutation was detected in NOTCH1. The significance of the mutation in this tumor is uncertain, since either activation or inactivation of NOTCH1 may be pathogenic, depending on the specific type of cancer.

NOTCH1 or notch homolog 1, translocation-associated, encodes a member of the Notch signaling network, an evolutionary conserved pathway that regulates developmental processes by regulating interactions between physically adjacent cells. Mutations in NOTCH1 play a central role in disruption of micro environmental communication, potentially leading to cancer progression. Due to the dual, bi-directional signaling of NOTCH1, activating mutations have been found in acute lymphoblastic leukemia and chronic lymphocytic leukemia, however loss of function mutations in NOTCH1 are prevalent in 11-15% of head and neck squamous cell carcinoma. NOTCH1 mutations have also been found in 2% of glioblastomas, 1% of ovarian cancers, 10% lung adenocarcinomas, 8% of squamous cell lung cancers and 5% of breast cancers. Notch pathway-directed therapy approaches differ depending on whether the tumor harbors gain or loss of function mutations, thus are classified as Notch pathway inhibitors or activators, respectively.

Additional Next-Generation Sequencing results continued on the next page. >

TN25-150648

PHYSICIAN: REEMA A. PATEL, MD

Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
ROS1	DNA-Tumor	Variant of Uncertain Significance	p.R2096L	40	c.6287_6288 delinsTT	41	NM_002944.2

Interpretation: A variant with no known clinical or functional significance was detected in ROS1.

The proto-oncogene ROS1 is a receptor tyrosine kinase of the insulin receptor family. The ligand and function of ROS1 are unknown. Dimerization of ROS1-fused proteins results in constitutive activation of the receptor kinase, leading to cell proliferation and survival.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
SMARCA4	DNA-Tumor	Pathogenic Variant	p.E1526*	33	c.4576G>T	68	NM_001128844.1

Interpretation: A loss of function pathogenic nonsense mutation was found.

The protein encoded by this gene is a member of the SWI/SNF family of proteins and is similar to the brahma protein of *Drosophila*. Members of this family have helicase and ATPase activities and are thought to regulate transcription of certain genes by altering the chromatin structure around those genes. In addition, this protein can bind BRCA1, as well as regulate the expression of the tumorigenic protein CD44. Mutations in this gene cause rhabdoid tumor predisposition syndrome type 2.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
TP53	DNA-Tumor	Pathogenic Variant	p.G117fs	4	c.348dupT	76	NM_000546.5

Interpretation: A pathogenic frameshift mutation was detected in TP53.

TP53, or p53, plays a central role in modulating response to cellular stress through transcriptional regulation of genes involved in cell-cycle arrest, DNA repair, apoptosis, and senescence. Inactivation of the p53 pathway is essential for the formation of the majority of human tumors. Mutation in p53 (TP53) remains one of the most commonly described genetic events in human neoplasia, estimated to occur in 30-50% of all cancers. Generally, presence of a disruptive p53 mutation is associated with a poor prognosis in all types of cancers, and diminished sensitivity to radiation and chemotherapy. Germline p53 mutations are associated with the Li-Fraumeni syndrome (LFS) which may lead to early-onset of several forms of cancer currently known to occur in the syndrome, including sarcomas of the bone and soft tissues, carcinomas of the breast and adrenal cortex (hereditary adrenocortical carcinoma), brain tumors and acute leukemias.

Additional Next-Generation Sequencing results continued on the next page. >

PATIENT:

TN25-150648

PHYSICIAN: REEMA A. PATEL, MD

Mutational Analysis by Next-Generation Sequencing (NGS)

GENES TESTED WITH INDETERMINATE* RESULTS BY TUMOR DNA SEQUENCING

ATP6AP2	EGLN1	MDH2	PREX2	RRAS2	TRRAP
B2M	EIF1AX	MDM2	PRKD1	SMARCA2	WRN
CDC73	EXO1	MRE11	PRKDC	SMARCE1	YES1
CDK6	FANCL	NPM1	RABL3	SOS1	
COL2A1	FBXW7	PLCB4	RAD50	SSBP1	
CUL3	FUBP1	PMS1	RASA1	STAG2	
DGCR8	JAK2	POLD3	REST	SUZ12	
EED	KIF1B	PRDM6	RHEB	TRIM28	

* Genes in this table were ruled indeterminate due to low coverage for some or all exons.

For a complete list of genes tested, visit www.CarisMolecularIntelligence.com/profilemenu.

NGS Methods

Direct sequence analysis was performed on genomic DNA isolated from a micro dissected tumor sample using Illumina NovaSeq 6000 sequencers. A hybrid pull down panel of baits was used to enrich more than 700 clinically relevant genes along with > 20,000 other genes. Sequence data is analyzed using a customized bioinformatics pipeline to detect sequencing variants, copy number alterations (amplifications and deletions) indels and HLA genotypes. In addition, genomic signatures for tumor mutational burden (TMB), microsatellite instability (MSI), genomic loss of heterozygosity (LOH) or HRD Genomic Scar Score (HRD GSS), and homologous recombination deficiency (HRD) are reported when applicable. For a complete list of what is covered by the assay, and genes with partial coverage, please contact Caris Customer Support.

Copy Number Alterations by Next-Generation Sequencing (NGS)

GENES TESTED WITH AMPLIFICATION DETECTED

ERBB2 (Her2/Neu)					
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GENES TESTED WITH INTERMEDIATE CNA RESULTS

GATA3					
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GENES TESTED WITH DELETION DETECTED

BCL2	SMAD4				
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CNA Methods

The copy number alteration (CNA) of each exon is determined by a calculation using the average sequencing depth of the sample along with the sequencing depth of each exon and comparing this calculated result to a pre-calibrated value. A complete list of genes for reporting copy number alterations, including amplifications and deletions, is available upon request.

Gene Fusion and Transcript Variant Detection by RNA Sequencing

Whole Transcriptome Sequencing (WTS) Methods

Gene fusion and variant transcript detection were performed on RNA isolated from a tumor sample using next generation sequencing. The assay also detects fusions occurring at known and novel breakpoints within genes. The genes included in this report represent the subset of genes associated with cancer. The complete list of unclassified alterations is available by request.

PATIENT:**TN25-150648****PHYSICIAN:** REEMA A. PATEL, MD

Protein Expression by Immunohistochemistry (IHC)

Biomarker	Patient Tumor			Thresholds
	Staining Intensity (0, 1+, 2+, 3+)	Percent of cells	Result	Conditions for a Positive Result:
ALK	0	100	Negative	Intensity $\geq 3+$ and $\geq 1\%$ of cells stained
MET	2+	5	Negative	Intensity $\geq 3+$ and $\geq 25\%$ of cells stained

PD-L1 TUMOR CELL STAINING				
Biomarker	Patient Tumor			Thresholds
	Staining Intensity (0, 1+, 2+, 3+)	Percent of cells	Result	Conditions for a Positive Result:
PD-L1 (28-8)	2+	100%	Positive	Intensity $\geq 1+$ and $\geq 1\%$ of cells stained
PD-L1 (SP142)	1+	50%	Positive	Intensity $\geq 1+$ and $\geq 50\%$ of cells stained
PD-L1 (SP263)	2+	100%	Positive	Intensity $\geq 1+$ and $\geq 1\%$ of cells stained

PD L1 (28 8): Scoring was based on percentage of viable tumor cells showing partial or complete membrane staining at any intensity.

PD L1 (SP142): TC scoring was based on the presence of discernible PD L1 membrane staining of any intensity in $\geq 50\%$ of viable tumor cells.

PD L1 (SP263): Scoring was based on percentage of viable tumor cells showing partial or complete membrane staining at any intensity.

PD-L1 TUMOR PROPORTION SCORE (TPS)			
Biomarker	Result	TPS	Threshold
PD-L1 (22c3)	Positive	100%	TPS $\geq 1\%$

PD L1 22c3: Scoring was based on the percentage of viable tumor cells showing partial or complete membrane staining. There are three categories of PD L1 expression defined by the PD L1 22c3 IHC pharmDx NSCLC interpretation guide: TPS < 1% (negative), TPS $\geq 1\%$ and TPS $\geq 50\%$.

Her-2 IHC: Biopsy	
Final Score	Threshold for Positive Result
Positive Score 3+	Intensity =3+ and ≥ 5 tumor cells staining

PD-L1 IMMUNE CELL (IC) SCORE			
Biomarker	Result	IC	Threshold
PD-L1 (SP142)	Negative	0%	$\geq 10\%$

PD L1 (SP142): IC scoring was based on discernible PD L1 staining of any intensity in tumor infiltrating immune cells covering $\geq 10\%$ of tumor area occupied by tumor cells, associated intratumoral or contiguous peritumoral stroma.

Clones used: ALK (D5F3), ERBB2 (Her2/Neu) (4B5), MET (SP44) RxDx Assay, PD L1 (22c3), PD L1 (28 8), PD L1 (SP142), PD L1 (SP263).



Electronic Signature

Amy Adams, MD

04/02/2025

Additional IHC results continued on the next page. >

TN25-150648

PHYSICIAN: REEMA A. PATEL, MD

Protein Expression by Immunohistochemistry (IHC)

IHC Methods

The Laboratory Developed Tests (LDT) immunohistochemistry (IHC) assays were developed and their performance characteristics determined by Caris Life Sciences. These tests have not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not currently necessary. Interpretations of all immunohistochemistry (IHC) assays were performed manually or with the assistance of an AI based image analysis tool by a board certified pathologist using a microscope and/or digital whole slide image(s).

The following IHC assays were performed using FDA approved companion diagnostic or FDA cleared tests consistent with the manufacturer's instructions: ALK (VENTANA ALK (D5F3) CDx Assay, Ventana), ER (CONFIRM anti Estrogen Receptor (ER) (SP1), Ventana), FOLR1 (VENTANA FOLR1 2.1 RxDx, Ventana), CLDN18 (VENTANA, 43 14A RxDx Assay), PR (CONFIRM anti Progesterone Receptor (PR) (1E2), Ventana), HER2/neu (PATHWAY anti HER 2/neu (4B5), Ventana), Ki 67 (MIB 1 pharmDx, Dako), MAGE A4 1F9 (pharmDx, Dako), PD L1 22c3 (pharmDx, Dako), PD L1 SP142 (VENTANA, non small cell lung cancer), PD L1 28 8 (pharmDx, Dako, gastric / GEJ, non small cell lung cancer), PD L1 SP263 (Ventana, non small cell lung cancer), and Mismatch Repair (MMR) proteins (MLH1, MSH2, MSH6, and PMS2; VENTANA MMR RxDx Panel, Ventana).

HER2 results and interpretation follow the ASCO/CAP scoring criteria. Bartley, A.N., J.A. Ajani, et al. (2016). "HER2 testing and clinical decision making in gastroesophageal adenocarcinoma: guideline from the College of American Pathologists, American Society for Clinical Pathology, and the American Society of Clinical Oncology". J Clin Oncol. 35(4):446 464.

References

#	Drug	Biomarker	Reference
1	alectinib	ALK	Camidge, D.R., A.T. Shaw, et al. (2019). "Updated Efficacy and Safety Data and Impact of the EML4-ALK Fusion Variant on the Efficacy of Alectinib in Untreated ALK-Positive Advanced Non-Small Cell Lung Cancer in the Global Phase III ALEX Study." J Thorac Oncol 14(7): 1233-1243. View Citation Online
2	alectinib	ALK	Gadgeel, S., D.R. Camidge, et al. (2018). "Alectinib versus crizotinib in treatment-naïve anaplastic lymphoma kinase-positive (ALK+) non-small-cell lung cancer: CNS efficacy results from the ALEX study." Ann Oncol 29 (11): 2214-2222. View Citation Online
3	alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	ALK	Lindeman, N.I., Y. Yatabe, et al. (2018). "Updated Molecular Testing Guideline for the Selection of Lung Cancer Patients for Treatment With Targeted Tyrosine Kinase Inhibitors Guideline From the College of American Pathologists, the International Association for the Study of Lung Cancer, and the Association for Molecular Pathology." J Thorac Oncol 13(3): 323-358. View Citation Online
4	alectinib, brigatinib, ceritinib, crizotinib, lorlatinib	ALK	National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology. Non-Small Cell Lung Cancer Version 1.2020
5	brigatinib	ALK	Camidge, D.R., S. Popat, et al. (2018). "Brigatinib versus Crizotinib in ALK-Positive Non-Small-Cell Lung Cancer." N Engl J Med 379(21): 2027-2039. View Citation Online
6	brigatinib	ALK	Lin, J.L., G.J. Riely, et al. (2018). "Brigatinib in Patients with Alectinib-refractory ALK-positive NSCLC." J Thorac Onc 13(10): 1530-1538. View Citation Online
7	brigatinib	ALK	Reckamp, K., J. Lee, et al. (2019). "Comparative efficacy of brigatinib versus ceritinib and alectinib in patients with crizotinib-refractory anaplastic lymphoma kinase-positive non-small cell lung cancer." Curr Med Res Opin 35(4):569-576. View Citation Online
8	brigatinib, crizotinib	ALK	Thorne-Nuzzo, T., P. Towne, et al. (2017). "A Sensitive ALK Immunohistochemistry Companion Diagnostic Test Identifies Patients Eligible for Treatment with Crizotinib." J Thorac Oncol 12(5): 804-813. View Citation Online
9	ceritinib, lorlatinib	ALK	Shaw, A.T., E. Felip, et al. (2017). "Ceritinib versus chemotherapy in patients with ALK-rearranged non-small-cell lung cancer previously given chemotherapy and crizotinib (ASCEND-5): a randomised, controlled, open-label, phase 3 trial." Lancet Oncol 18 (7):874-886. View Citation Online
10	ceritinib, lorlatinib	ALK	Solomon, B. J., A. T. Shaw, et al. (2018). "Lorlatinib in patients with ALK-positive non-small cell lung cancer: results from a global phase 2 study." Lancet Oncol 19:1654-1667. View Citation Online
11	ceritinib, lorlatinib	ALK	Soria, J.C., de Castro G., et al. (2017). "First-line ceritinib versus platinum-based chemotherapy in advanced ALK-rearranged non-small-cell lung cancer (ASCEND-4): a randomised, open-label, phase 3 study." Lancet 389: 917-929. View Citation Online
12	crizotinib	ALK	Solomon, B. J., T. S. Mok, et al. (2018). "Final overall survival analysis from a study comparing first-line crizotinib versus chemotherapy in ALK-mutation positive Non-small-cell-lung cancer." J Clin Oncol 36: 2251-2258. View Citation Online
13	crizotinib	ALK	van der Wekken, A. J., H.J.M. Groen, et al. (2017). "Dichotomous ALK IHC is a better predictor for ALK inhibition outcome than traditional ALK FISH in advanced Non-small cell lung cancer." Clin Cancer Res 23(15): 4251-4258. View Citation Online
14	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Hyman, D.H., J. Baselga, et al. (2015). "Vemurafenib in Multiple Nonmelanoma Cancers with BRAF V600 Mutations." NEJM 373(8):726-736. View Citation Online
15	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Planchard, D., B.E. Johnson, et al. (2016). "An open-label phase II trial of dabrafenib (D) in combination with trametinib (T) in patients (pts) with previously treated BRAF V600E mutant advanced non-small cell lung cancer (NSCLC; BR113928)." J Clin Oncol 34: 15_suppl, 107-107. View Citation Online
16	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Planchard, D., B.E. Johnson, et al. (2017). "Dabrafenib plus trametinib in patients with previously untreated BRAFV600E-mutant metastatic non-small-cell lung cancer: an open-label, phase 2 trial." Lancet Oncol 18(1):1307-1316. View Citation Online

References

#	Drug	Biomarker	Reference
17	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Riely, G. J., B. E. Johnson, et al. (2023). "Phase II, Open-Label Study of Encorafenib Plus Binimetinib in Patients With BRAFV600-Mutant Metastatic Non-Small-Cell Lung Cancer." J Clin Oncol. 41 (21): 3700-3711. View Citation Online
18	dabrafenib, dabrafenib + trametinib, encorafenib + binimetinib, vemurafenib	BRAF	Subbiah, V., D.M. Hyman, et al. (2019). "Efficacy of Vemurafenib in Patients With Non-Small-Cell Lung Cancer With BRAF V600 Mutation: An Open-Label, Single-Arm Cohort of the Histology-Independent VE-BASKET Study." JCO Precis Oncol 2019; 3, 1-9. View Citation Online
19	afatinib, dacomitinib, osimertinib	EGFR	Sequist, L.V., M. Schuler, et al. (2013). "Phase III Study of Afatinib or Cisplatin Plus Pemetrexed in Patients with Metastatic Lung Adenocarcinoma With EGFR Mutations." J Clin Oncol ahead of print July 1, 2013, doi: 10.1200/JCO.2012.44.2806. View Citation Online
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